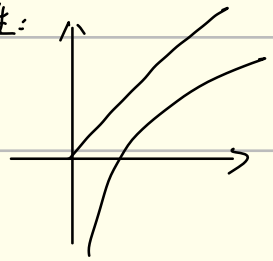
判断级数敛散性:

1. $\sum_{n=2}^{\infty} \frac{1}{\sqrt[n]{n}}$



(等价、阶的判断、几种判别法.

如果无思路，比较判

可以证得

$\sqrt{n} > \ln n \quad (n>4)$

$\Rightarrow n > \ln^2 n$

$\frac{1}{n} < \frac{1}{\ln^2 n}$ 而 $\sum_{n=2}^{\infty} \frac{1}{n}$ 发散，所以 $\sum_{n=2}^{\infty} \frac{1}{\sqrt[n]{n}}$ 发散.

2. $\sum_{n=1}^{\infty} \frac{1}{n!}$

$\lim_{n\to+\infty} \frac{u_{n+1}}{u_n} = \frac{1}{n+1} = 0 < 1 \quad \therefore \sum_{n=1}^{\infty} \frac{1}{n!}$ 收敛.

3. $\sum_{n=1}^{\infty} \frac{\ln n}{n^2}$

考虑P级数 $\sum_{n=1}^{\infty} \frac{1}{n^{3/2}}$

$\lim_{n\to+\infty} \frac{\ln n \cdot n^{3/2}}{n^2} = \lim_{n\to+\infty} \frac{\ln n}{\sqrt{n}} = \lim_{n\to+\infty} \frac{\frac{1}{n}}{\frac{1}{2\sqrt{n}}} = \frac{2\sqrt{n}}{n} = \frac{2}{\sqrt{n}} = 0. \quad (n\to+\infty).$

故. 收敛.

4. $\sum_{n=1}^{\infty} (\sqrt[n]{n}-1)$,

$u_n = n^{\frac{1}{n}} - 1 = e^{\frac{1}{n}\ln n} - 1$

$\lim_{n\to+\infty} \frac{\ln n}{n} = 0. \quad \therefore e^{\frac{1}{n}\ln n} - 1 \sim \frac{\ln n}{n} \quad (n\to+\infty).$

$\frac{\ln n}{n} > \frac{1}{n}. \quad \therefore$ 发散.

6. $\sum_{n=1}^{\infty} \frac{n^2}{\sqrt[n]{n^2}}$ $\quad \because \sqrt[n]{n^2} \quad \frac{\sqrt[n]{n^2}}{\quad} = \frac{1}{\quad} < 1$

5. $\sum_{n=1} 2^n$ $\lim_{n\to+\infty} \sqrt{u_n} = \sqrt{2^n} = \lim_{n\to+\infty} 2 = 2 < 1$

收敛.

6. $\sum_{n=1}^{\infty} \frac{[2+(-1)^n]^n}{2^{2n+1}}$

$u_n < \frac{3^n}{2^{2n+1}} = \frac{1}{2} \cdot \frac{3^n}{4^n}$

↑

收敛.

故收敛.

7. $\sum_{n=1}^{\infty} n^2 e^{-n}$

$\lim_{n\to+\infty} \sqrt[n]{n^2 e^{-n}} = \lim_{n\to+\infty} \sqrt[n]{n^2} \cdot \lim_{n\to+\infty} \sqrt[n]{e^{-n}} = \frac{1}{e} < 1$

收敛.

8. $\sum_{n=1}^{\infty} \frac{2^n n!}{n^n}$

$\lim_{n\to+\infty} \frac{2^{n+1}(n+1)!}{(n+1)^{n+1}} \cdot \frac{n^n}{2^n n!} = \frac{2(n+1)\cdot n^n}{(n+1)^{n+1}} = \frac{2\cdot n^n}{(n+1)^n} = 2 > 1$

发散.

9. $\sum_{n=1}^{\infty} (\sqrt{n^2+1} - \sqrt{n^2-1})$ ☆ 启示：可以判断它的阶 $\frac{1}{n}, \frac{1}{n^2}$ ...

$u_n = \sqrt{n^2+1} - \sqrt{n^2-1}$ $n^2+1-(n^2-1)$

$= \frac{(\sqrt{n^2+1} - \sqrt{n^2-1})(\sqrt{n^2+1} + \sqrt{n^2-1})}{\sqrt{n^2+1} + \sqrt{n^2-1}} = \frac{2}{\sqrt{n^2+1} + \sqrt{n^2-1}} \sim \frac{2}{n}$

$\lim \frac{\sqrt{n^2+1} + \sqrt{n^2-1}}{2\sqrt{n^2}} = \lim_{n\to+\infty} \sqrt{1+\frac{1}{n^2}} + \lim_{n\to+\infty} \sqrt{1-\frac{1}{n^2}}$

$= 2$

发散.

10. $\sum_{n=1}^{\infty}\left(2n-\sqrt{n^2+1}-\sqrt{n^2-1}\right)$

考虑 P级数 $\frac{1}{n^2}$

$$\lim_{n\to+\infty}\frac{2n-\sqrt{n^2+1}-\sqrt{n^2-1}}{n^2}=0.$$

$\sum_{n=1}^{\infty}\frac{1}{n^2}$ 收敛 $\therefore$ 此级数收敛.

11. $\sum_{n=2}^{\infty}\ln\frac{n^2+1}{n^2-1}$

$u_n=\ln\frac{n^2+1}{n^2-1}=\ln\left(\frac{n^2-1+2}{n^2-1}\right)$

$=\ln\left(1+\frac{2}{n^2-1}\right).$

$(n\to+\infty)$ $u_n\sim\frac{2}{n^2-1}\sim\frac{2}{n^2}$

收敛.

12. $\sum_{n=3}^{\infty}\left(-\ln\cos\frac{\pi}{n}\right).$

$u_n=-\ln\cos\frac{\pi}{n}$

$=-\ln\left(1+\cos\frac{\pi}{n}-1\right)$

$=-\cos\frac{\pi}{n}+1\quad(n\to+\infty).$

$=1-\cos\frac{\pi}{n}.$

$1-2\sin^2x=\cos 2x.$

$= 2\sin^2\frac{\pi}{2n} \le 2\cdot\frac{\pi}{4n^2}$ $\quad$ $1-\cos 2x = 2\sin^2 x$

收敛.

13. $\sum_{n=1}^{\infty}\left(a^{\frac{1}{n^2}}-1\right)$.

$U_n = a^{\frac{1}{n^2}}-1 = e^{\frac{1}{n^2}\ln a}-1 \sim \frac{\ln a}{n^2}$ $\quad (n\to+\infty)$.

$\frac{\ln a}{n^2} \sim \frac{1}{n^2}$ 收敛.

14. $\sum_{n=1}^{\infty} 2^n\cdot\sin\frac{\pi}{3^n}$

$U_n = 2^n\cdot\sin\frac{\pi}{3^n} \le 2^n\cdot\frac{\pi}{3^n} = \pi\cdot\left(\frac{2}{3}\right)^n$ 收敛.

15. $\sum_{n=1}^{\infty}\frac{1}{1+a^n}$ $(a>0)$.

$U_n = \frac{1}{1+a^n}$

若 $a<1$. $\quad a^n\to 0$, $U_n\to 1$. 发散

$a>1$ $\quad \lim_{n\to+\infty}\frac{\frac{1}{1+a^n}}{\frac{1}{a^n}} = \lim_{n\to+\infty}\frac{a^n}{1+a^n} = \frac{1}{1+\frac{1}{a^n}} = 1$.

$a>1$, 则 $\sum\frac{1}{a^n}$ 收敛

$\therefore$ 收斂.

16. $\sum_{n=1}^{\infty} \frac{\ln n}{n^{4/3}}$ $\quad u_n = \frac{\ln n}{n^{4/3}}$ 收斂.